

Maxwell's Demon: The 0/0 of the Fine Lens

Michael Grafiel S Puno

2026

The Removable Singularity Framework

Abstract

Ch.67 measured the arrow: it lives in the coarse lens. Maxwell (1867) proposed the one who refuses the lens; Szilard (1929) turned him into an engine; Landauer (1961) and Bennett (1982) found his price. A Szilard engine is simulated for 100,000 rounds per error rate: (1) measured mean work obeys $E[W] = k_B T \ln 2 (1-2p)$ to ~1% - at $p=0$ it is exactly $k_B T \ln 2 = 2.87098e-21$ J (ratio 1.000000); a blind demon ($p=0.5$) extracts ~0. (2) The value of one bit of information is $k_B T \ln 2$ - information and heat are the same coin. (3) The demon's memory stores one bit per round; erasure of the 100,000-bit bank costs $2.87098e-16$ J and the measured net ledger is $1.69e-28$ J: zero to $5.9e-13$ relative. The demon is paid exactly what it earned; there is no perpetual motion. The two entropies exchange at the rate $k_B \ln 2$ per bit: the 0/0 of entropy. The lens is paid; the arrow is the account book that balances.

2. The Szilard Engine (1929)

$$E[W] = k_B T \ln 2 * (1 - 2p) \quad \text{measured to } \sim 1\%$$

One particle, half a box: measure the half (p = error rate), insert a piston, expand isothermally. Work per round is $+k_B T \ln 2$ when the guess is right, $-k_B T \ln 2$ when wrong. Measured over 100,000 rounds: $p=0.00 \rightarrow 2.87098e-21$ J (ratio 1.000000); $p=0.10 \rightarrow 2.3035e-21$; $p=0.25 \rightarrow 1.4319e-21$; $p=0.45 \rightarrow 2.92e-22$. The linear law holds to simulation noise. The work comes from the INFORMATION, not from any energy gradient.

3. The Value of the Bit

The knowing engine ($p=0$) earns $k_B T \ln 2$ per round; the blind engine ($p=0.5$) earns $\sim 3e-24$ J ~ 0 . The difference is exactly $k_B T \ln 2 = 2.87098e-21$ J: one bit of information has a measured price, and it is the price Ch.59 paid as the heat of creation. One bit refunds one nat of the lens (Ch.67).

4. The Bank (Landauer 1961, Bennett 1982)

$$\text{Work } N * k_B T \ln 2 - \text{Erasure } N * k_B T \ln 2 = 0 \text{ (measured)}$$

The demon's memory holds one bit per round; erasing the bank of 100,000 bits costs $100,000 * k_B T \ln 2 = 2.87098e-16$ J. Work extracted: $2.87098e-16$ J. Net ledger: $1.69e-28$ J - zero to $5.9e-13$ relative. Done reversibly (Ch.65) the erasure could cost arbitrarily little, but the book must be cleared; the Second Law is the account book that balances.

5. The 0/0 Proof

Thermodynamic entropy and information entropy exchange at the rate $k_B \ln 2$ per bit: it is the 0/0 of entropy, the joint of the two currencies (Ch.59's heat of creation is this same rate). The demon is the Fine Lens personified (Ch.67): what coarsening spent as heat, the demon refunds; the bank re-bills it. The lens is paid; the arrow is the ledger.

6. Connections to Prior 0/0 Singularities

Arrow of time (Ch.48): the demon reverses the arrow. Landauer (Ch.59): the heat of creation is the bill. Reversible cycle (Ch.65): reversible erasure costs ~0. The lens (Ch.67): the demon is the lens's advocate. Meaning (Ch.61): bits are the currency of the ledger. What the light throws away, the eye finds - and pays for.

References

- [1] Maxwell (1867), letter to Tait: the sorting demon**
- [2] Szilard (1929), On the decrease of entropy in a thermodynamic system**
- [3] Landauer (1961), Irreversibility and heat generation in the computing process**
- [4] Bennett (1982), The thermodynamics of computation - a review**
- [5] Loyd (1982), After the demon (information-theoretic Szilard)**
- [6] Puno, "The Removable Singularity" (2026), chapters 48-67**